Jainta Paul

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Summary

Former Software Engineer - Pursuing Ph.D. in Computer Science. Built static-analysis platforms, automated security pipelines, and program-analysis tools, and led end-to-end development of systems involving OpenPLC/LLVM instrumentation, machine-learning evaluation pipelines, and agent-based cyber-physical experimentation. Proficient in Python, Java, C++, and JavaScript, with strong experience in software development, automation, security, and systems engineering.

Education

University of Utah, Kahlert School of Computing *Jan 2024 – Present*
Ph.D. Student in Computer Science, Ongoing Salt Lake City, UT

University of Utah, Kahlert School of Computing *May 2026*
M.S. in Computer Science, GPA: 4.00/4.00 Salt Lake City, UT

Bangladesh University of Engineering and Technology *Feb 2017 – May 2022*
B.Sc. in Computer Science and Engineering, GPA: 3.32/4.00 Dhaka, Bangladesh

Technical Skills

Languages: Python, Java, C++, JavaScript, SQL
Systems & Tools: Git, Docker, Linux, CUDA, PyTorch
Software/Security: Static Analysis, Taint Analysis, Vulnerability Detection, Threat Modeling

Software Engineering Experience

Software Engineer *Jun 2022 – Nov 2023*
OpenRefactory, Inc.

- **Test Drive – OpenRefactory:** Designed and led a GitHub-integrated web platform that allowed users to run static-analysis trials on real-world repositories, with OAuth authentication, repository selection, automated job orchestration, and and result reporting.
- **Open-Source Bug Discovery Pipeline:** Built automated workflows to analyze open-source repositories, identify security and compliance issues, triage findings, and generate reports for maintainers, reducing the amount of manual review needed in the vulnerability reporting process.
- **Intelligent Code Repair:** Improved the Java Intelligent Code Repair engine by implementing call-graph construction, points-to analysis, and static taint tracking, expanding its ability to detect vulnerabilities and generate more precise repairs across large codebases.

Selected Research Projects

ICSTRACKER: Intrusion Backtracking for Industrial Control Systems *DSN 2025*
Paper: IEEE Xplore **Code:** GitHub

- Developed a provenance-based intrusion backtracking framework for Industrial Control Systems (ICS) that reconstructs attack paths from physical process anomalies to their originating cyber events.
- Built and instrumented the MiniWater OpenPLC testbed, integrating C/C++, LLVM, and Python-based tracing to capture controller execution and runtime events.
- Designed data collection and analysis pipelines for generating structured execution traces to support cyber-physical attack attribution.

Sensor Privacy as a Spectrum: Quantifying Privacy in Edge and Multimodal Systems through Games *PETS 2026*
Paper: PoPETs **Code:** GitHub

- Led the development of a privacy evaluation framework for edge and on-sensor machine learning systems using adversarial inference attacks.
- Built data-processing pipelines and implemented multiple machine learning adversaries (Decision Trees, Random Forests, CNNs, LSTMs, and Transformers) to quantify information leakage from wearable sensor outputs.
- Evaluated privacy-utility tradeoffs across sensing modalities, output granularities, temporal contexts, and hardware configurations.

ICS-AGENT: Agent-Based Wargaming for Industrial Control Systems Cyber Security *2025 – Present*

- Led development of a cyber-physical wargaming framework that models attacker–defender interaction over an OpenPLC-based water-treatment testbed.
- Implemented telemetry collection, process-state tracking, attacker actions, defender mitigations, and safety-bound evaluation for closed-loop ICS experiments.
- Built scripted, rule-based, and LLM-driven agent workflows and evaluated them across benchmark attack scenarios using detection, mitigation, containment, and safety-violation metrics.

Towards Cross-Physical-Domain Threat Inference for Industrial Control System Defense Adaptation *ACM CCS-RICSS 2024*
Paper: ACM Digital Library

- Developed a threat inference framework that correlates cyber attacks with physical process behavior to improve adaptive defense in industrial control systems.
- Modeled cross-domain attack propagation using process semantics to support earlier detection and mitigation of cyber-physical threats.

HyTwin: A Digital Twin Framework for Hybrid Systems Verification *NASA Formal Methods (NFM) 2025*
Paper: Springer Nature

- Contributed to a digital twin framework for hybrid systems that combines runtime system observations with formal verification models.
- Assisted in implementing verification workflows for reasoning about cyber-physical system behavior under dynamic operating conditions.

Service & Awards

- Research Mentor, University of Utah REU, Summer 2024.
- Artifact Evaluator, USENIX VehicleSec 2026 and ACSAC 2025.
- Graduate Tuition & Research Scholarship, University of Utah; University Admission Scholarship, BUET.

References

Dr. Luis Garcia
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